Towards Understanding the Predictability of Stock Markets from the Perspective of Computational Complexity

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This paper initiates a study into the century-old issue of market predictability from the perspective of computational complexity. We develop a simple agent-based model for a stock market where the agents are traders equipped with simple trading strategies, and their trades together determine the stock prices. Computer simulations show that a basic case of this model is already capable of generating price graphs which are visually similar to the recent price movements of high tech stocks. In the general model, we prove that if there are a large number of traders but they employ a relatively small number of strategies, then there is a polynomial-time algorithm for predicting future price movements with high accuracy. On the other hand, if the number of strategies is large, market prediction becomes complete for two new computational complexity classes CPP and promise-BCPP, where $\mathrm{P}^{\mathrm{NP}[O(\log n)]} \subseteq \mathrm{BPP}_{\mathrm{path}} \subseteq$ promise-BCPP $\subseteq$ CPP $=$ PP. These computational hardness results open up a novel possibility that the price graph of an actual stock could be sufficiently deterministic for various prediction goals but appear random to all polynomial-time prediction algorithms.

1 Introduction

The issue of market predictability has been debated for more than a century (see [8] for earlier papers and [22,15,6,20] for more recent viewpoints). In 1900, the pioneering work "Theory of Speculation" of Louis Bachelier used Brownian motion to analyze the stochastic properties of security prices [8]. Since then, Brownian motion and its variants have become textbook tools for modeling financial assets. Relatively recently, the radically different methodology of Mandelbrot used fractals to approximate price graphs deterministically [23]. In this paper, we initiate a study into this long-running issue from the perspective of computational complexity.

We develop a simple agent-based model for a stock market [9,21]. The agents are traders equipped with simple trading strategies, and their trades together determine the stock prices. We first consider a basic case of this model where there are only two strategies, namely, momentum and contrarian strategies. The choice of this base model and thus our general model is justified at two levels: (1) Experimental and empirical studies in the finance literature [1,7,10,11,19,7,4,12] show that a large number of traders primarily

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follow these two strategies. (2) Our own simulation results show that despite its simplicity, the base model is capable of generating price graphs which are visually similar to the recent price movements of high tech stocks (Figs. 1 and 2).

With these justifications, we then consider the issue of market predictability in the general model. We prove that if there are a large number of traders but they employ a relatively small number of strategies, then there is a polynomial-time algorithm to predict future price movements with high accuracy (Theorem 1). On the other hand, if there are also a large number of strategies, then the problem of predicting future prices becomes computationally very hard. To describe this hardness, we define two new computational complexity classes called CPP and promise-BCPP (Definitions 1 and 4). We show that some market prediction problems are hard for these two classes (Theorems 5 and 6) and that $\mathrm{P}^{\mathrm{NP}[O(\log n)]} \subseteq \mathrm{BPP}_{\mathrm{path}} \subseteq$ promise-BCPP $\subseteq$ CPP $=$ PP.

These computational completeness results open up the possibility that the price graph of an actual stock could be sufficiently deterministic for various prediction purposes but appear random to all polynomial-time prediction algorithms. This is in contrast to the most popular academic belief that the future price of a stock cannot be predicted from its historical prices because the latter are statistically random and contain no information. This new possibility also differs from the fractal-based methodology in that the price graph of a stock could be a fractal but the fractal might not be computable in polynomial time. The findings in this paper can by no means settle the debate about market predictability. Our goal is only that our alternative approach could provide new insights to the predictability issue in a systematic manner. In particular, it could provide a general framework to investigate the many documented technical trading rules [25] and to generate novel and significant interdisciplinary research problems for computer science and finance.

The rest of the paper is organized as follows. Section 2 discusses the basic market model. Section 3 formulates the general model. Section 4 proves the complexity results for market prediction in the general model. We conclude the paper with some directions for future research in Sect. 5.

2 A Basic Market Model

In this section, we present a very simple market model, called the *deterministic-switching MC* (DSMC) model. The letter M stands for a *momentum* strategy, and the letter C for a *contrarian* strategy. These two strategies and the model itself are defined in Sect. 2.1. Some computer simulations for this model are reported in Sect. 2.2.

Intuitively, these strategies are heuristics ("rules of thumb") used by traders in the absence of reliable asset valuation models. As discussed in [12], a momentum trader may observe a sequence of "up" trades (price incre-